LESSON 3.10

Convert Measurement Units

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can convert measurement units using ratios and conversion factors.

WHY THIS MATTERS You're building a brand-new arcade cabinet! The blueprints use feet, but the parts catalog lists measurements in inches. The wiring instructions use yards, and the weight limits are in pounds — but the shipping labels use ounces. You need to convert between units to make sure everything fits and ships correctly. The blueprint says the cabinet is 6 feet tall, and the parts catalog lists a side panel that is 70 inches tall. Does the panel fit?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion Factor *Spanish: Factor de conversión*	A ratio that helps you change one unit into another.	1 foot = 12 inches, so the factor is 12
Unit *Spanish: Unidad*	A standard amount used to measure, like an inch or a liter.	Inches, feet, pounds, ounces
Equivalent *Spanish: Equivalente*	Having the same value, just in different units.	2 feet = 24 inches
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	12 inches per 1 foot
Customary units *Spanish: Unidades usuales (sistema inglés)*	The U.S. way of measuring, like inches, feet, and pounds.	12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1 pound
Metric units *Spanish: Unidades métricas*	A way of measuring by 10s, like meters, liters, and grams.	1 meter = 100 centimeters, 1 kilogram = 1,000 grams

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The cabinet is 5 feet tall. I want to know how many inches that is.

1. My conversion factor is 1 foot = 12 inches.
2. Inches are smaller than feet, so I multiply: 5 × 12 = 60.
3. So 5 feet = 60 inches.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A wire is 4 yards long. The factor is 1 yard = 3 feet.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When converting from a _____ unit to a _____ unit, I _____ by the conversion factor because _____.

Word bank Conversion Factor | Unit | Equivalent | Ratio | Customary units | Metric units | larger | smaller | multiply | divide | conversion factor | ratio

WATCH OUT!

A common mistake

A common mistake in Convert Measurement Units is applying the conversion factor in the wrong direction

LESSON 3.10

Convert Measurement Units

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

How many inches are in 7 feet?

(A) 70 inches
(B) 72 inches
(C) 84 inches
(D) 96 inches

STEPS TO TRY

1. The key fact: 1 foot = 12 inches. You have 7 feet and want inches.
2. Going from a BIGGER unit (feet) to a SMALLER unit (inches) means you multiply.
3. Set up 7 × 12 — seven groups of 12 inches.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

How many feet are in 3 yards?

(A) 6 feet
(B) 9 feet
(C) 12 feet
(D) 36 feet

STEPS TO TRY

1. The key fact: 1 yard = 3 feet. You have 3 yards and want feet.
2. Big unit (yards) to small unit (feet) means multiply by the conversion factor 3.
3. Set up 3 × 3 — three yards, each worth 3 feet.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many ounces are in 4 pounds?

(A) 16 ounces
(B) 32 ounces
(C) 48 ounces
(D) 64 ounces

STEPS TO TRY

1. The key fact: 1 pound = 16 ounces. You have 4 pounds and want ounces.
2. Pounds are bigger than ounces, so converting to ounces means multiplying by 16.
3. Set up 4 × 16. (Break it up: 4 × 10 = 40 and 4 × 6 = 24, then add.)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Convert 24 inches to feet.

(A) 2 feet
(B) 3 feet
(C) 12 feet
(D) 24 feet

STEPS TO TRY

1. This one goes the OTHER direction: inches (small) to feet (big), so the number of units gets smaller.
2. Small unit to big unit means divide. Ask: how many groups of 12 inches fit in 24 inches?
3. Set up 24 ÷ 12.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

SENTENCE STARTER When converting from a ___ unit to a ___ unit, I ___ by the conversion factor because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help